6.1 The common-source (CS) stage of Fig. 6.1 is designed with $(W/L)_1$ = 50/0.5, R_S = 100 Ω, and R_D = 2 KΩ. If I_{D1} is biased to 1 mA, C_{GS}=20fF, C_{GD}=0.2fF, C_{DB}=20fF, $\mu_n C_{OX}$=1.34×10^{-4} A/V^2, V_{DD}=3V, V_{TH}=0.7V, and λ=0, please answer the following questions:

(a) Based on Eqn. (6.30) in the textbook obtained through direct analysis of the small-signal model, determine the and the zero and poles of the circuit by solving the roots of the numerator and denominator of the transfer function, respectively.

(b) Estimate the zero and poles and using Miller approximation, and compare the results with those obtained in (a).

(c) Write down the overall transfer function of the amplifier, and sketch the frequency response curves of the amplifier for questions (a) and (b) in a single figure, including both magnitude (in dB) and phase responses. The ω axis is required to be shown in logarithmic scale (log(ω).

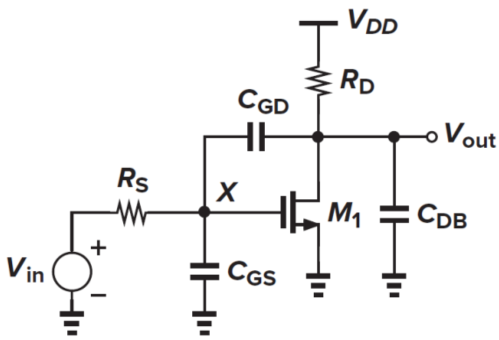


Fig. 6.1

Solution: (a)

$$\frac{V_{out}}{V_{in}} = \frac{(C_{GD}s - g_m)R_D}{R_S R_D \xi s^2 + [R_S(1+g_m R_D)C_{GD} + R_S C_{GS} + R_D(C_{GD}+C_{DB})]s + 1}$$

$$\xi = C_{GS}C_{GD} + C_{GS}C_{DB} + C_{GD}C_{DB}$$

$$g_m = \sqrt{2I\mu_n C_{OX}(W/L)} = 5.2\text{mS}$$

Substitute the parameters into 6.30, we can obtain:

$\omega_z = g_m/C_{GD}$ = 2.588×10^{13} rad/s

$\omega_{p1,da}$ = -2.46×10^{10} rad/s

$\omega_{p2,da}$ = -4.98×10^{11} rad/s

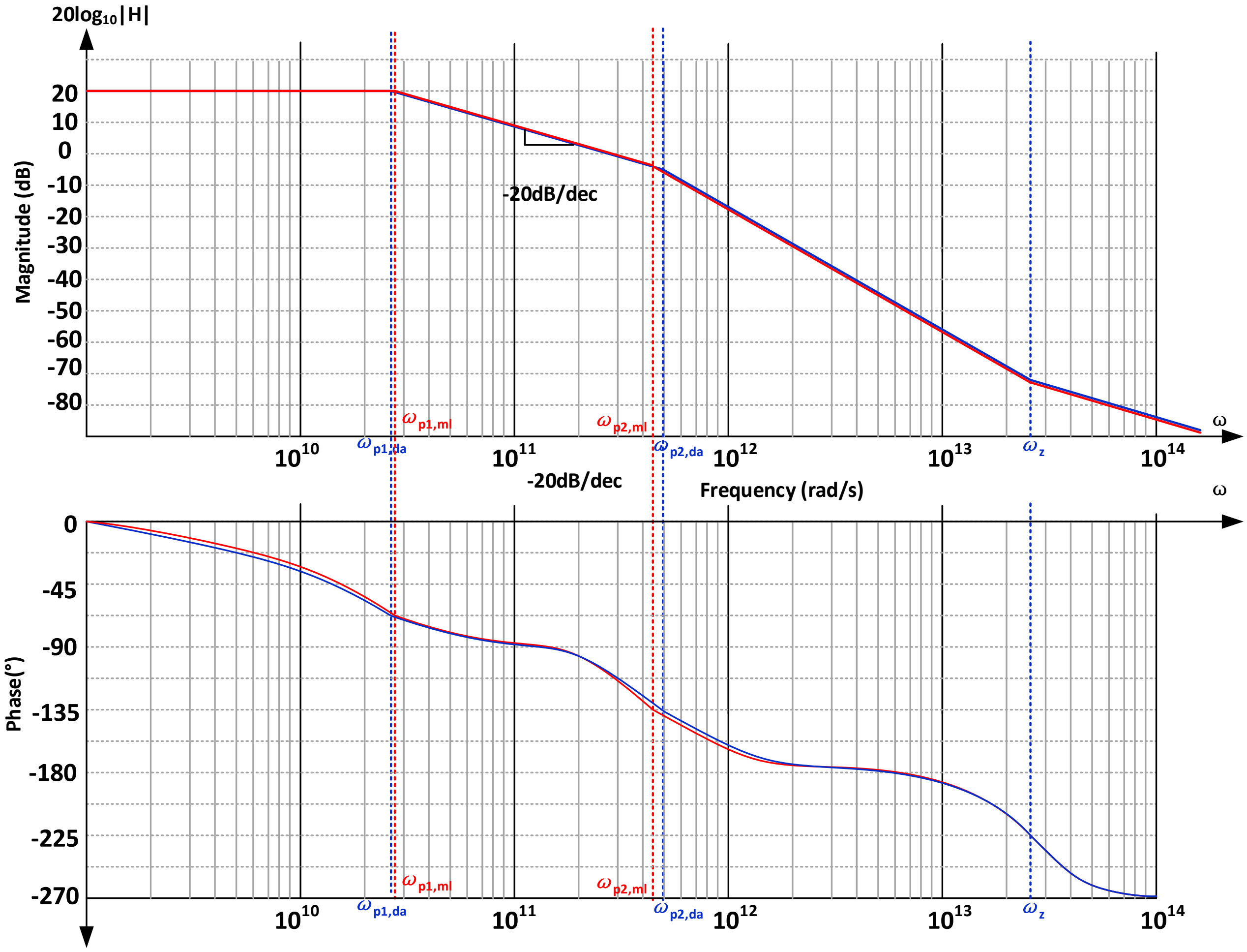
(b) As both RS and CGD are relatively small, the frequencies of ω_{p1} and ω_{p2} can be approximated as:

$$\omega_{p2,ml} = -\frac{1}{((1+g_m R_D)C_{GD} + C_{GS})R_S} = -4.49\times10^{11} \text{ rad/s}$$

$$\omega_{p1,ml} = -\frac{1}{(C_{GD}+C_{DB})R_D} = -2.475\times10^{10} \text{ rad/s}$$

(c) The low-frequency gain is

$A_{V0} = -g_m R_D =$ –10.35, corresponding to a magnitude of 20*log10(10.35) = 20.3dB, therefore the frequency response can be plotted as



6.2 The class-AB amplifier in Fig. 6.2 is designed with $(W/L)_1$ = 50/0.5, $(W/L)_2$ = 100/0.5. The DC current of the amplifier is biased to 1 mA, and both M_1 and M_2 are operating in saturation region. The process parameters are $\mu_n C_{OX}$=1.34×10^{-4}A/V^2, $\mu_p C_{OX}$=0.38×10^{-4}A/V^2, V_{THN}=0.7V, V_{THP}=–0.8V, $\lambda_N=\lambda_P$=1/(22V). Other circuit parameters are V_{DD}=3V and R_S=100KΩ. Considering only the 3 capacitors in the circuit, C_{IN}=20fF, C_C=400fF, C_L=600fF, please answer the following questions:

(a) Find the low-frequency gain of the amplifier.

(b) Based on Eqn. (6.30) in the textbook, determine the and the zero and poles of the circuit by solving the roots of the numerator and denominator with the circuit parameters, respectively.

(c) Estimate the zero and poles and using Miller approximation, and compare the results with those obtained in (b).

(d) Write the overall transfer function of the amplifier for question (c), and sketch the frequency response curves, including both amplitude (in dB) and phase responses. The ω axis is required to be shown in logarithmic scale.

(e) Calculate the gain-bandwidth product and unit-gain frequency based on the results from question (c).

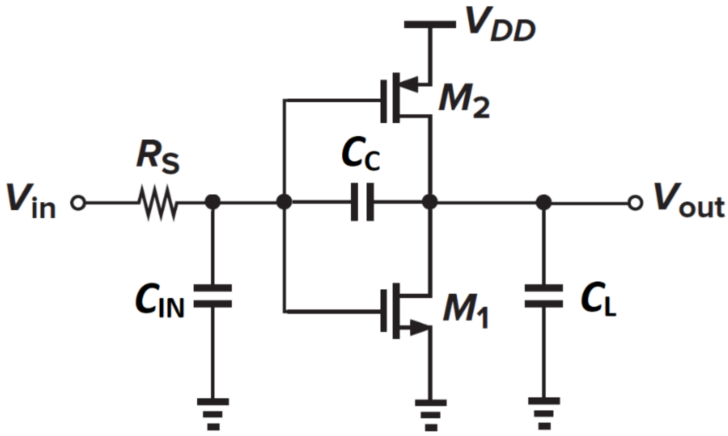


Fig. 6.2

Solution: (a)

$$g_{m,n}=\sqrt{2I\mu_n C_{OX}\left(W/L\right)_n}\quad = 5.2\text{mS}$$

$$g_{m,p}=\sqrt{2I\mu_p C_{OX}\left(W/L\right)_p}\ =3.9\text{mS}$$

$$g_m = g_{m,p}+g_{m,n}\ =9.1\text{mS}$$

$$r_{oN}=r_{oP}=1/(\lambda I_D)=220\text{ k}\Omega \rightarrow R_{out}=R_D=r_{oN}||r_{oP}=110\text{ k}\Omega$$

C_{IN}=20fF, C_C=400fF, C_L=600fF –> C_{GS}=20fF, C_{GD}=400fF, C_{DB}=600fF

$A_{V0}=-g_m\times R_{out}\approx 100$

(b)

$$\frac{V_{out}}{V_{in}}=\frac{(C_{GD}s-g_m)R_D}{R_S R_D \xi s^2+[R_S(1+g_m R_D)C_{GD}+R_S C_{GS}+R_D(C_{GD}+C_{DB})]s+1}$$

$$\xi = C_{GS}C_{GD}+C_{GS}C_{DB}+C_{GD}C_{DB}$$

Substitute the parameters into 6.30, we can obtain:

$\omega_z = g_m/C_{GD} = 2.27\times10^{10}$ rad/s

$\omega_{p1,da} = -2.47\times10^{5}$ rad/s

$\omega_{p2,da} = -2.09\times10^{10}$ rad/s

(c) As both RS and CGD are relatively small, the frequencies of ω_{p1} and ω_{p2} can be approximated as:

$$\omega_{p1,ml} = -\frac{1}{((1+g_m R_D)C_{GD} + C_{GS})R_S} = -2.48\times10^{5} \text{ rad/s}$$

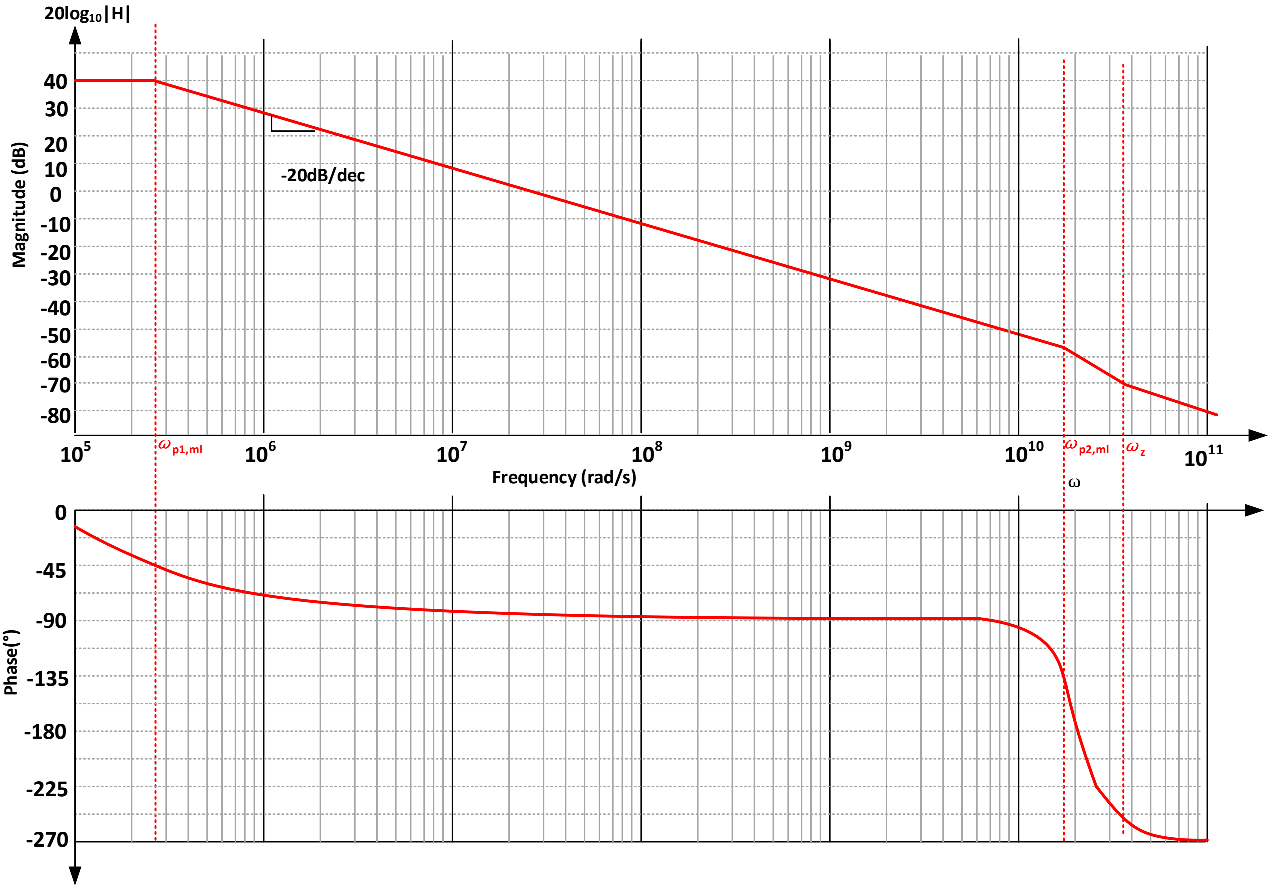
$$\omega_{p2,ml} = -\frac{g_m}{C_{GS} + C_{DB}} = -2.16\times10^{10} \text{ rad/s}$$

$\omega_z = g_m/C_{GD} = 2.27\times10^{10}$ rad/s

(d) The low-frequency gain is

$A_{V0} = -g_m R_D = -100$, corresponding to a magnitude of 20*log10(100) = 40dB,

therefore the frequency response can be plotted as



(e)GBW = $|A_V\times\omega_{p1}|=2.47\times10^{7}$ rad/s.

As GBW<< ω p2, $\omega_u \approx GBW = 2.47\times10^{7}$ rad/s

6.3 Calculate the gain of each circuit in Fig. 6.3 at very low and very high frequencies. Neglect all other capacitances and assume that $\lambda = \gamma = 0$. (HINT: capacitors can be thought as open and short at very low and very high frequencies, respectively)

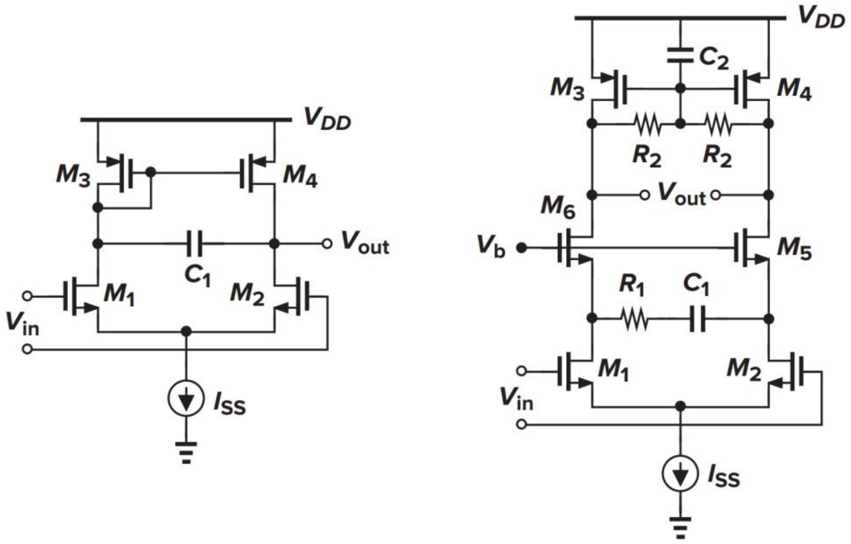
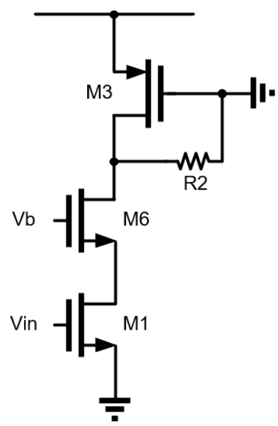


Fig. 6.3

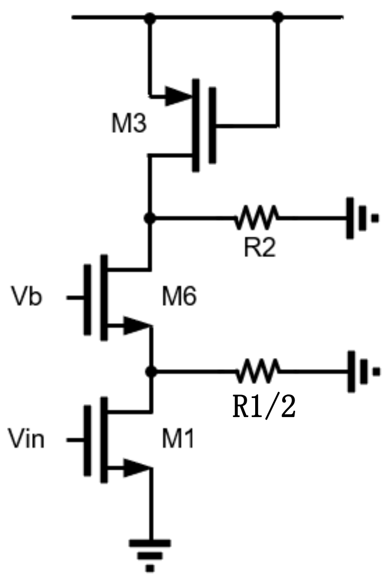
(1)At very low frequencies, the left picture can be seen as 5-transistor OTA.

So $A_v = g_{m1}(r_{02} // r_{o4}) = \infty$

At very high frequencies, C_1 in the left picture can be replaced with a line. So there is no differential gain.
$A_v = 0$
(2)At very low frequencies, half of the right picture can be seen as follows:



$R_{out} = g_{m6}(r_{o6} r_{o1}) // R_2 // r_{o3}$ $\quad A_v = -g_{m1} R_{out}$
At very high frequencies, half of the right picture can be seen as follows:



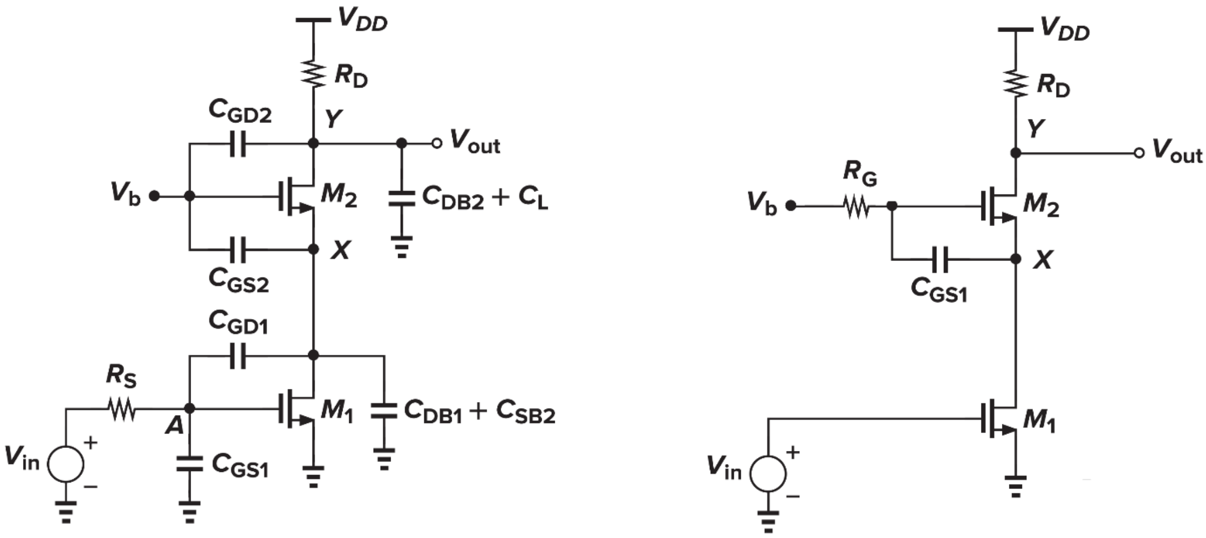
We can easily get $R_{out}=R_2 // r_{o3} // (g_{m6}r_{o6}(r_{o1} // (R_1/2))$

$A_v=-g_{m1}R_{out}$

6.4 For the circuits given in Fig. 6.4, if M1 and M2 are operating in saturation, $\lambda = \gamma = 0$, answer the following questions:

(a) If $g_{m1}=4g_{m2}$, derive the transfer function of the circuit in Fig. 6.4 (a);

(b) Suppose a resistor RG appears in series with the gate of M2, as shown in Fig. 6.4(b), considering only C_{GS2} (neglecting other capacitances), derive the transfer function.



(a) (b)

Fig. 6.4

Solution:

(a) Refer to the lecture notes, re-derive it;

$$第一级放大倍数 A_{v1} = -\frac{g_{m1}}{g_{m2}} = -4$$

$$则 \omega_A = \frac{1}{(C_{GS1} + 5C_{GD1})R_S}$$

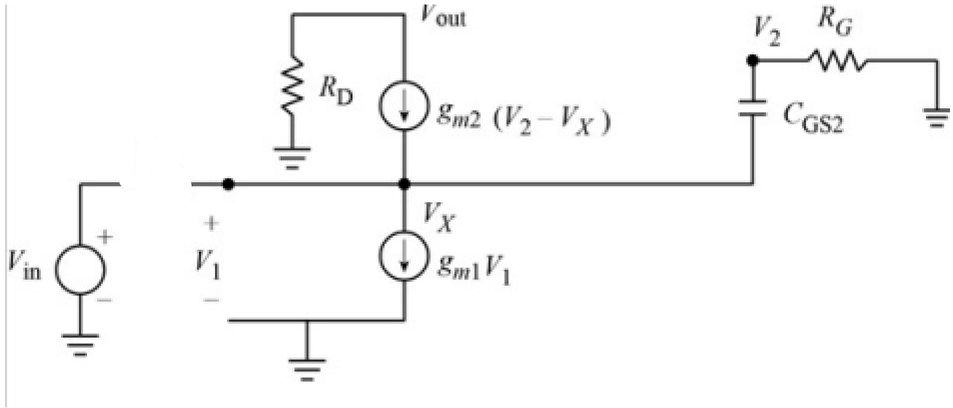
$$\omega_X = \frac{g_{m2}}{\frac{5}{4}C_{GD1} + C_{DB1} + C_{SB2} + C_{GS2}}$$

$$\omega_Y = \frac{1}{R_D(C_{GD2} + C_{DB2} + C_L)}$$

$$第二级放大倍数 A_{v2} = g_{m2}R_D$$

$$故 \frac{V_{out}}{V_{in}} = -\frac{4g_{m2}R_D}{\left(1+\frac{\omega}{\omega_A}\right)\left(1+\frac{\omega}{\omega_X}\right)\left(1+\frac{\omega}{\omega_Y}\right)}$$

(b)画出小信号模型如图所示



我们有

$$\begin{cases} V_1 = V_{in} \\ (V_2 - V_x)g_{m2} + (V_2 - V_x)sC_{GS2} = g_{m1}V_1 \\ V_{out} = -g_{m2}(V_2 - V_x) \end{cases}$$

最后可得 $\frac{V_{out}}{V_{in}} = -\frac{g_{m1}g_{m2}R_D}{g_{m2} + sC_{GS}}$

6.5The circuit of Fig. 6.5 is designed with (W/L)1,2 = 50/0.5 and (W/L)3,4 = 10/0.5，(W/L)5,6=50/0.5. The VB is a DC bias voltage. The process parameters are μnCOX=1.34×10-4A/V2 , μpCOX=0.38×10-4A/V2 , VTHN=0.7V, VTHP=– 0.8V, γ=0, λ=0.1V-1. Other circuit parameters are ISS = 100 μA, K = 2,VDD=3V, and all MOS are operating in saturation. Consider only the capacitors in the figure: CGD1= CGD2=20fF, CX= CY=366fF, CL=42.28fF.

(a) Calculate the low-frequency differential gain of the amplifier. (HINT: For simplicity, you can let λ=0 to calculate ID and gm of the transistors, while use λ=0.1V-1 to calculate their ro only).

(b) Assuming the amplifier is driven by an ideal differential voltage source (RS=0), estimate the differential-mode poles and zeros of the circuit on the basis of (a).

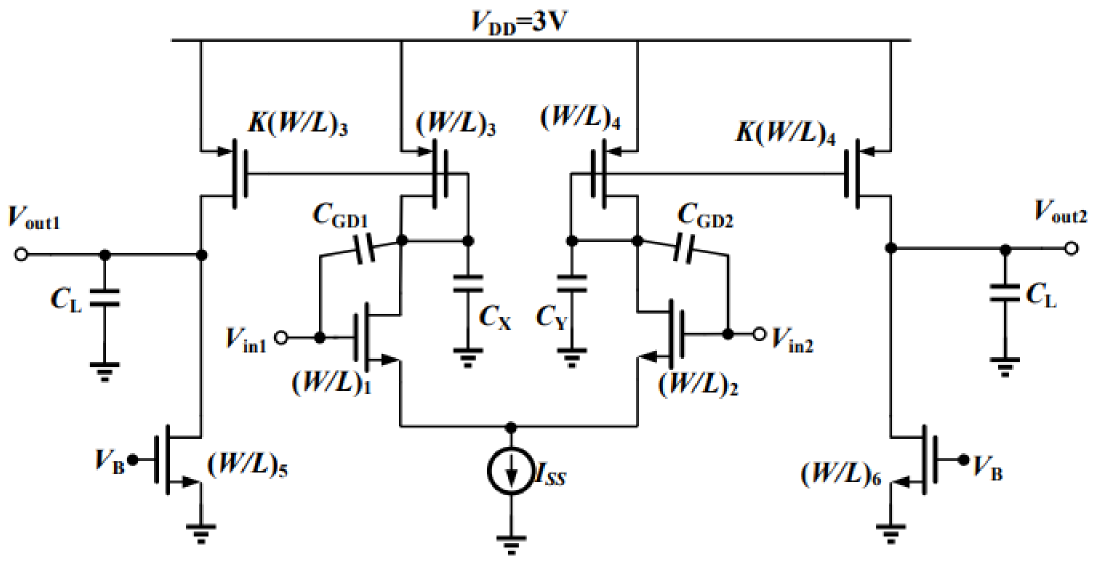
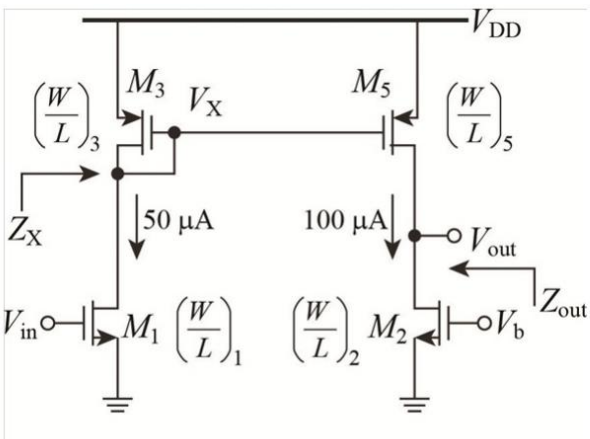


Fig. 6.5

（a）

$$A_v = \frac{g_{m1}}{g_{m3}} K g_{m3} \frac{r_{o5}}{2} = K\frac{g_{m1} r_{o5}}{2} = 115.8$$

（b）可以采用半边电路等效法进行分析，获得电路如下：



零点产生：零点的产生原因是第一级放大器中 V_{in} 在电容 C_{GD1} 产生的电流和在 M1 中产生电流形成回路，不对输出贡献，此时有:$sC_{GD1}V_{in}=g_{m1}V_{in}$，此时可以得到 $s_0=g_{m1}/C_{GD1}$

$$g_{m1} = \sqrt{I_{SS}\mu_n C_{ox}\left(\frac{W}{L}\right)_1} = 1.158mS$$

$$s_0 = 5.75 \times 10^{10} rad \cdot s^{-1}$$

对于 X 点的极点，我们有如下分析:

$$\omega_X = -\frac{1}{((1 - A_{v1}{}^{-1})C_{GD} + C_X)\frac{1}{g_{m3}}}$$

$$A_{v1} = -\frac{g_{m1}}{g_{m3}}$$

$$g_{m3} = \sqrt{I_{SS}\mu_p C_{ox}\left(\frac{W}{L}\right)_3} = 2.757 \times 10^{-4} S$$

$$\text{解得}\omega_X = -7.055 \times 10^8 rad \cdot s^{-1}$$

对于输出点的极点，我们有如下分析:

$$\omega_L = -\frac{1}{(r_{o2} \parallel r_{o5})C_L}$$

$$r_{o2} = r_{o5} = \frac{1}{\lambda I_{ss}} = 10^5 \Omega$$

$$\text{解得}\omega_X = -4.73 \times 10^8 rad \cdot s^{-1}$$

6.6 A differential pair driven by an ideal voltage source is required to have a total phase shift of 135° at the frequency where its gain drops to unity.

(a) Explain why a topology in which the load is realized by diode-connected devices or current sources does not satisfy this condition.

(b) Consider the circuit shown in Fig. 6.6. Neglecting other capacitances, determine the transfer function. Explain under what conditions the load exhibits an inductive behavior. Can this circuit provide a total phase shift of 135° at the frequency where its gain drops to unity?

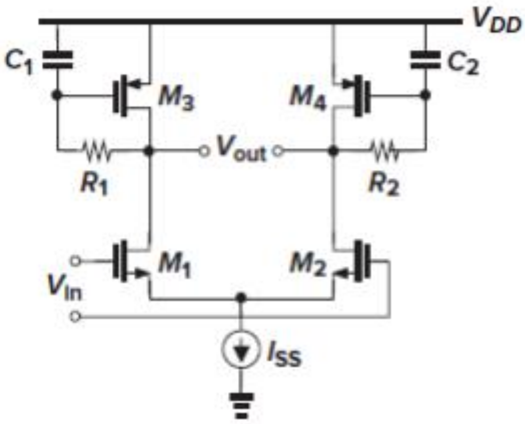


Fig. 6.6

(a)

以二极管连接为例子进行分析,

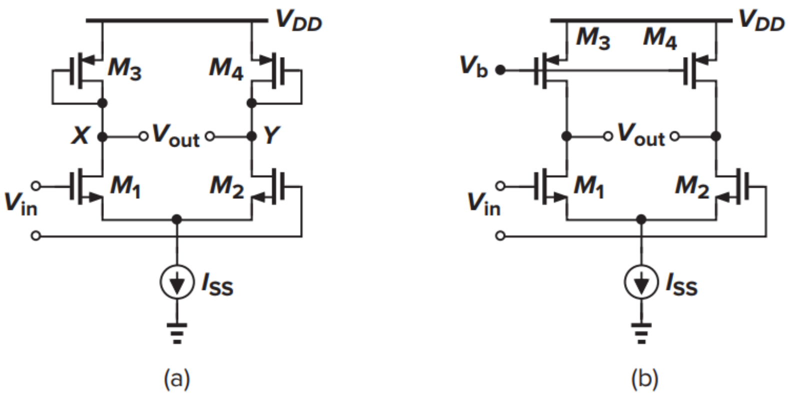
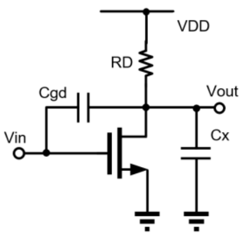


Figure 4.37 Differential pair with (a) diode-connected and (b) current-source loads.

我们画出下图（a）的半边电路



我们有:

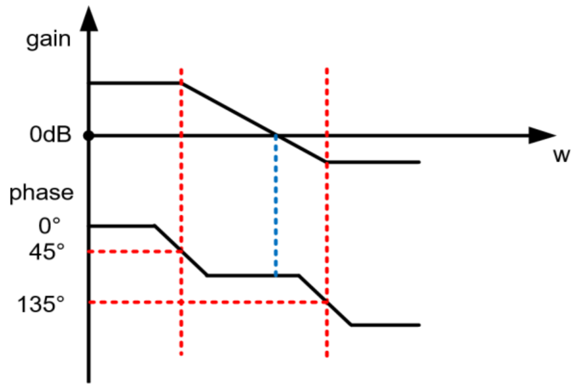
$$R_D=\frac{1}{g_{m3}}，\quad C_X=2C_{DB1}+C_{DB3}+C_{GB3}$$

$$\frac{V_{out}}{V_{in}}=\frac{(sC_{GD1}-g_{m1})\frac{1}{g_{m3}}}{1+s\frac{1}{g_{m3}}C_X}$$

我们可以得到$z_0=\dfrac{g_{m1}}{C_{GD1}},\omega_X=-\dfrac{g_{m3}}{C_X},$

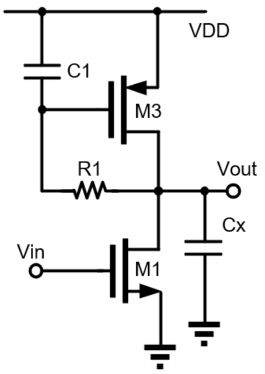
故$|z_0|\gg|\omega_X|$

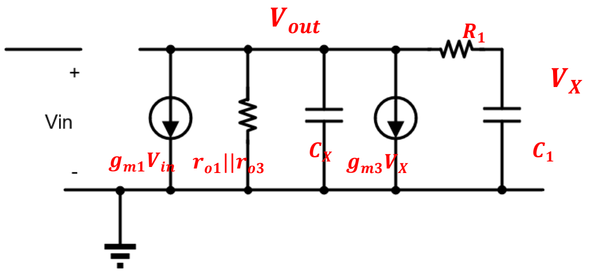
该系统有一个右零点和一个左极点，我们得到下图，容易得出，不可能在单位增益时有 135°相移



电流源接法的分析和二极管接法相同。

(b) 为获得传递函数，我们画出小信号模型：





对于X：$\dfrac{V_{out}-V_X}{R_1}=sC_1V_X$

对于节点OUT：$g_{m1}+\dfrac{V_{out}}{r_{o1}\,/\!/\,r_{o3}}+sC_XV_{OUT}+g_{m3}V_X+\dfrac{V_{out}-V_X}{R_1}=0$

故$\dfrac{V_{OUT}}{V_X}=\dfrac{g_{m1}(1+sR_1C_1)}{s^2C_XR_1C_1+s\left(C_X+C_1+\frac{R_1C_1}{r_{o1}\,/\!/\,r_{o3}}\right)+g_{m3}+\frac{1}{r_{o1}\,/\!/\,r_{o3}}}$

$\omega_z=-\dfrac{1}{R_1C_1}\qquad \omega_{p1}+\omega_{p2}=-\dfrac{b}{a}=-\dfrac{1}{R_1C_1}-\dfrac{1}{R_1C_X}-\dfrac{1}{C_X(r_{o1}\,/\!/\,r_{o3})}$

我们可以得到$|\omega_{p1}|+|\omega_{p2}|>2|\omega_z|$

所以只有以下两种情况：

$|\omega_{p1}|>|\omega_z|\geqslant|\omega_{p2}|$或者$|\omega_{p1}|>|\omega_{p2}|>|\omega_z|$

无论是哪种情况，都不会先让两个极点同时起主要作用，零点会最先或者在第一个极点起主要作用后抬高相位，所以不可能出现 135° 的相移。只有零点晚于两个极点出现时才有可能发生。

因为电感是阻交流的，所以只要输出电阻很大的时候，我们就认为是感性的，显然在高频时，输出电阻要远高于低频时的二极管连接，故高频时可以看为感性的。